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Decision Analytics for Sustainable E-Commerce Logistics: A Multi-Criteria Framework for Evaluating Delivery and Return Policies

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Abstract: E-commerce retailers increasingly face the challenge of balancing customer-oriented delivery incentives with logistics efficiency and economic sustainability. Delivery and return policies therefore constitute a multi-criteria decision problem involving competing operational, financial, and customer-related objectives. This study investigates how alternative delivery and return policies can be evaluated within a structured decision-analytics framework for sustainable e-commerce logistics. An integrated Method based on the Removal Effects of Criteria (MEREC)-Multi-Objective Optimization on the basis of Simple Ratio Analysis (MOOSRA) framework was developed, in which MEREC was used to derive objective criterion weights and MOOSRA was applied to rank five policy alternatives. The evaluation was based on a modeled dataset representing a small domestic e-commerce retailer during increased sales activity from November 2025 to January 2026. The results showed that customer-paid delivery achieved the highest performance score (1.590), followed by threshold-based free delivery combined with a prepayment incentive (1.096) and threshold-based free delivery with enhanced return management (1.073), whereas unconditional free delivery ranked last (0.341). Sensitivity analysis further showed that the first-ranked alternative remained unchanged across all tested perturbations of the scenario-based assessment matrix and criterion weights. The findings indicate that higher order activity does not necessarily translate into superior economic performance when logistics costs and return-related risks are considered. The proposed framework provides a transparent decision-support approach for evaluating economically viable and resource-efficient e-commerce logistics policies under multiple competing criteria.

Keywords: E-commerce logistics; Sustainable logistics; Decision analytics; Multi-criteria decision-making; Delivery policy; Return management; Method based on the Removal Effects of Criteria; Multi-Objective Optimization on the basis of Simple Ratio Analysis

1 Introduction

In recent years, e-commerce has become an increasingly established channel for purchasing goods, allowing consumers to search for products, compare prices, and place orders within minutes without visiting a physical store. Behind this apparent simplicity, however, lies a sequence of closely interconnected activities that must be coordinated effectively. Once an order is placed, its details must be verified, inventory must be allocated, the shipment must be prepared and transferred to a delivery provider, and its subsequent progress must be monitored. Potential returns and customer complaints must also be managed. Failures at any stage of this process may have consequences extending beyond warehouse or transportation operations because, from the customer's perspective, the retailer remains responsible for the overall purchasing experience regardless of whether a particular problem originates with the retailer, the delivery provider, or another participant in the supply chain [1–3].

This development has strengthened the relationship between marketing and logistics. In traditional retail, customers can directly inspect a product, communicate with sales personnel, and usually take possession of the purchased item immediately. E-commerce separates the purchase decision from physical receipt of the product, creating a period during which customers depend on the retailer and its logistics partners for order confirmation, accurate information on expected delivery, shipment visibility, and fulfillment reliability. Upon delivery, customers expect the received product to correspond to its description and arrive in satisfactory condition. When dissatisfaction

occurs, they also expect transparent information regarding return procedures and the conditions under which products can be returned [4, 5]. Logistics therefore constitutes not merely a supporting operational function but an integral part of the value proposition offered by an online retailer.

The quality of logistics services can influence both customer response and business performance. Customers effectively purchase not only a physical product but also an expectation that it will be delivered within an acceptable timeframe, in the expected condition, and under clearly communicated terms. Reliable fulfillment can support customer satisfaction and repeat purchasing, whereas delivery failures may generate complaints, failed transactions, reverse-logistics costs, and negative word-of-mouth. Recent industry evidence reinforces the importance of these issues. DHL's 2025 e-commerce research indicates that a substantial proportion of consumers abandon purchases when preferred delivery or return options are unavailable and that trust in delivery providers remains an important element of online purchasing decisions [6, 7]. These findings suggest that consumers evaluate not only product characteristics and prices but also delivery costs, expected delivery times, available delivery locations, and return conditions. Similarly, Kawa and Zdrenka [8] identified relationships among logistics value, customer satisfaction, customer loyalty, and the performance of online retailers.

Returns create an additional layer of complexity. A simple and convenient return process can reduce customers' perceived purchasing risk and support sales, but it also generates transportation, inspection, handling, repackaging, refund-processing, and potential inventory-value costs. Returned products may need to be restored to inventory, repackaged, refurbished, discounted, or written off depending on their condition. Recent retail evidence further demonstrates that product returns represent a substantial and recurring component of contemporary e-commerce operations rather than an exceptional event [9]. Delivery incentives create a comparable trade-off. Free delivery may increase conversion or encourage higher order values, yet it transfers transportation costs from customers to retailers and can weaken the contribution generated by individual transactions. Threshold-based delivery policies may mitigate this effect by encouraging larger baskets, but their effectiveness depends on product margins, logistics costs, customer behavior, and the risk of unsuccessful deliveries. Consequently, delivery and return policies should not be assessed using a single indicator such as order volume, conversion rate, or delivery cost. Their evaluation requires simultaneous consideration of customer response, logistics performance, operational risk, and financial outcomes.

Within this context, delivery and return policies constitute a structured multi-criteria decision problem involving competing customer-oriented, operational, logistics-risk, and financial objectives. A policy that stimulates order creation may simultaneously increase delivery expenditure, unclaimed shipments, or product returns, while a more restrictive policy may reduce logistics exposure at the cost of lower customer attractiveness. The central analytical problem is therefore not simply whether a particular policy increases sales, but whether the resulting balance of customer response, retained sales, logistics risk, and financial contribution is preferable to that produced by competing alternatives [10]. This perspective is particularly relevant to sustainable e-commerce logistics because economically sustainable service policies must reconcile customer expectations with the retailer's capacity to maintain reliable and financially viable fulfillment operations.

Accordingly, this study aims to develop and apply a multi-criteria decision-analytics framework for evaluating alternative delivery and return policies in sustainable e-commerce logistics. More specifically, the study seeks to identify a set of customer-related, operational, logistics-risk, and financial criteria relevant to policy evaluation; determine their relative importance through an objective weighting procedure; rank alternative delivery and return policy scenarios; and examine the trade-offs underlying the resulting priorities. The analysis therefore treats logistics not solely as a cost-generating operational function but as a factor that can influence customer behavior, realized sales, and overall business performance. Efficiently designed logistics processes may facilitate purchasing decisions, reduce perceived purchase risk, increase average order value, and encourage repeat purchases, whereas inadequately designed free-delivery or return policies may stimulate apparent demand while reducing the profitability of retained order rate.

To operationalize this decision problem, the study integrates the Method based on the Removal Effects of Criteria (MEREC) and Multi-Objective Optimization on the basis of Simple Ratio Analysis (MOOSRA) methods. MEREC is employed to determine objective criterion weights by examining how the removal of individual criteria changes the overall performance assessment of the alternatives. Criteria whose exclusion produces greater changes in the aggregated evaluation receive greater importance, thereby reflecting their influence on the decision structure. MOOSRA is subsequently used to rank the policy alternatives on the basis of the weighted criteria. The method provides a ratio-based mechanism for aggregating benefit- and cost-oriented performance dimensions and has been reported to offer comparatively stable rankings without generating negative performance scores, particularly in comparison with the conventional MOORA formulation [11]. The integration of these methods provides a transparent structure for evaluating competing delivery and return policies across heterogeneous criteria rather than relying on isolated indicators or managerial intuition.

The analytical application is based on a modeled dataset representing the operations of a small domestic e-

commerce retailer during a period of increased sales activity from November 15, 2025, to January 15, 2026. The dataset was constructed using realistic business parameters consistent with the characteristics of the analyzed online retail model, although the resulting values do not represent data obtained from a specific company. The selected period covers the weeks surrounding the New Year holidays, when website traffic and order volumes may increase while delivery providers simultaneously face greater operational workloads. These conditions provide an appropriate setting for examining logistics-related challenges such as delivery delays, unclaimed shipments, damaged parcels, and product returns, as well as the financial consequences associated with alternative delivery policies. Particular attention is given to retained order rate, defined as successfully delivered orders that remain sales after subsequent returns, because this measure provides a more realistic representation of realized business performance than created orders alone.

The contribution of the study lies in bringing delivery incentives and return management into a unified decision structure rather than considering them as separate operational issues. By jointly incorporating customer response, retained-order performance, logistics risks, implementation requirements, and financial contribution, the proposed framework makes explicit the trade-offs that underlie alternative e-commerce logistics policies. It also distinguishes among created orders, dispatched orders, successfully delivered orders, and retained order rate, thereby incorporating the economic consequences of unclaimed shipments and post-delivery returns into policy evaluation. In methodological terms, the study demonstrates how objective criterion weighting and multi-criteria ranking can be integrated into a transparent decision-support framework for sustainability-oriented logistics management. In the present study, sustainability is interpreted primarily through the economic viability and resource efficiency of e-commerce logistics. Particular attention is given to avoidable fulfillment failures, unsuccessful deliveries, reverse-logistics flows, repeated handling, and logistics expenditure that does not result in retained order rate.

The remainder of the paper is organized as follows. Section 2 presents the research problem and reviews the relevant literature. Section 3 describes the principal processes of distribution logistics and reverse logistics. Section 4 provides a practical analysis of the effects of delivery and returns on sales and profitability. Section 5 develops the proposed decision model, including the MEREC-based criterion weighting procedure and MOOSRA-based ranking of delivery and return policy alternatives. Finally, Section 6 presents the conclusions, managerial implications, limitations, and directions for future research.

2 Problem Description and Literature Review

The relationship between marketing and logistics in e-commerce can be understood through the broader concept of customer value. Marketing is concerned not only with attracting customers and stimulating transactions but also with creating, communicating, and delivering value. From this perspective, the customer evaluates an offering according to both what is received and what must be sacrificed to obtain it, including monetary expenditure, time, convenience, and perceived risk. Kotler and Keller [12] emphasize the role of value creation and delivery in marketing, while Kotler and Armstrong [13] underline the importance of customer satisfaction and long-term customer relationships. In e-commerce, these considerations extend directly to logistics because the purchasing process is not completed at the point at which an online order is submitted. Product availability, order processing, delivery reliability, communication, complaint handling, and returns all contribute to the value ultimately experienced by the customer.

This connection between logistics and customer value is particularly important in online retail, where the customer cannot normally inspect the product before purchase and must rely on information provided by the retailer. Logistics therefore influences both the functional completion of the transaction and the perceived quality of the retailer's service. Christopher [14] links logistics performance with competitive advantage through two complementary objectives: increasing customer value and controlling total cost. This perspective is especially relevant to e-commerce because retailers frequently offer comparable products and compete not only through price but also through delivery reliability, fulfillment convenience, product availability, and the simplicity of return procedures. Christopher and Peck [15] further emphasize that competitive advantage emerges across the broader chain through which value is delivered rather than within isolated organizational functions. Consequently, marketing decisions cannot be separated from inventory availability, warehouse capacity, packaging, transportation, and delivery performance.

This interdependence creates an important managerial problem. Promotional activities may increase website traffic and the number of orders while simultaneously placing additional pressure on warehouse and delivery operations. If fulfillment capacity does not increase accordingly, apparently successful marketing activity may lead to delayed shipments, incorrect orders, failed deliveries, complaints, and returns. Evaluating the performance of an e-commerce campaign exclusively through the number of orders generated can therefore produce misleading conclusions. The number of successfully delivered orders, the proportion of unclaimed shipments, subsequent product returns, average order value, logistics costs, and ultimately retained order rate are also relevant to the assessment of business performance.

The customer perspective provides further evidence of this relationship. E-commerce logistics service quality is multidimensional rather than being determined by delivery speed alone. Akıl and Ungan [16] identified factors

such as timeliness, order condition, order accuracy, and discrepancy handling as important dimensions of customer satisfaction and loyalty. Customers may tolerate several days of delivery when the expected delivery period is communicated accurately and fulfilled consistently. By contrast, unpredictable delivery performance and inconsistent information may generate dissatisfaction even when the absolute delivery time is relatively short. The quality of information therefore constitutes part of logistics service quality together with the physical movement of products.

Kawa and Zdrenka [8] similarly demonstrated that logistics value is related to customer satisfaction, customer loyalty, and the performance of online retailers. Their findings reinforce the view that customers perceive the e-commerce offering as an integrated combination of product, price, delivery, communication, and return conditions. From the customer's perspective, a delay, damaged package, incorrect item, or difficult return procedure forms part of the experience with the retailer rather than representing an independent logistics event. This integrated perspective is important because positive logistics experiences can strengthen confidence in subsequent transactions, whereas repeated service failures can weaken customer relationships even when the products themselves meet expectations.

Recent industry evidence also indicates that delivery and return conditions influence online purchasing decisions. DHL's e-commerce studies report that customers may abandon purchases when suitable delivery or return options are unavailable and that confidence in delivery and return providers influences willingness to complete online transactions [6, 7]. Out-of-home delivery options, including parcel lockers and pickup locations, can provide additional flexibility while potentially reducing unsuccessful delivery attempts [17]. These observations indicate that logistics policies influence customer behavior before an order is placed as well as the operational and financial outcomes that follow the transaction.

Delivery pricing represents one of the clearest examples of this interaction. A retailer may require customers to pay the full delivery charge, absorb the delivery cost for all orders, or offer free delivery once an order reaches a specified value. Each option generates a different combination of customer attractiveness and financial exposure. A relatively high delivery charge may discourage purchases, particularly for inexpensive products. Unconditional free delivery may increase conversion but reduce the financial contribution generated by individual orders. A free-delivery threshold may encourage customers to increase basket value, although the effectiveness of such a policy depends on the relationship among the threshold, average order value, product margins, and logistics costs.

The way in which delivery charges are communicated is also relevant. If additional costs become visible only at the final stage of checkout, customers may perceive the offer as less transparent even when the total price remains competitive. The Serbian Law on Electronic Commerce requires information affecting the price to be disclosed clearly [18]. From a managerial perspective, early disclosure can also reduce unexpected changes in the final purchase price and enable customers to evaluate the complete offer before entering the final stage of the transaction.

Return policies create a comparable trade-off. Flexible return procedures can reduce perceived purchasing risk and increase customer confidence, yet they can also generate reverse-transportation, inspection, handling, repackaging, refund-processing, and inventory costs. A returned item may be immediately resalable, require repackaging or refurbishment, be sold at a reduced price, or become unsuitable for further sale. Consequently, the financial consequence of a return depends not only on the transportation cost but also on the condition and subsequent disposition of the returned product. Shopify's guidance stresses that return policies should be accessible, clear, and understandable to customers [19, 20]. From an operational perspective, however, return convenience must also be balanced against the costs and risks created by the policy.

The literature therefore identifies several closely related dimensions of e-commerce logistics performance. These include customer value, logistics service quality, delivery reliability, fulfillment transparency, delivery flexibility, return convenience, cost control, and customer satisfaction. However, the individual research streams tend to emphasize different parts of the problem. Marketing-oriented studies focus primarily on value creation and customer relationships, logistics studies emphasize service quality and operational performance, and practitioner-oriented sources focus on particular delivery or return practices. Table 1 synthesizes these perspectives and clarifies their relationship with the decision problem addressed in this study.

The synthesis in Table 1 shows that previous literature provides substantial evidence for the importance of individual components of e-commerce logistics but offers a less integrated treatment of the policy decision itself. Existing studies explain why delivery reliability matters, how logistics service quality influences satisfaction and loyalty, why return convenience affects customer confidence, and how logistics contributes to competitive advantage. These findings provide the conceptual basis for selecting relevant evaluation criteria. Nevertheless, they do not fully resolve the managerial problem of how alternative delivery and return policies should be compared when improvements in one dimension produce deterioration in another.

This limitation becomes particularly visible when marketing, operational, and financial indicators are examined together. A free-delivery policy, for example, may improve purchase conversion while increasing the transportation cost borne by the retailer. A more flexible return policy may strengthen customer confidence while increasing the probability and processing cost of product returns. Cash-on-delivery may reduce perceived purchasing risk for new customers but increase exposure to unclaimed shipments. Similarly, a promotional campaign may generate a large

number of submitted orders while producing a much smaller number of successfully delivered and retained order rate. These relationships reveal a decision structure characterized by competing criteria rather than a problem that can be resolved through a single performance indicator.

Table 1. Synthesis of literature relevant to delivery and return policy evaluation in e-commerce

Source	Main Focus	Principal Perspective	Relevance to Delivery and Return Decisions	Limitation Relative to the Present Study
Kotler and Keller [12]; Kotler and Armstrong [13]	Customer value, satisfaction, and long-term relationships	Marketing and customer management	Establishes that value depends on the complete customer offering rather than the product alone	Does not provide a formal framework for comparing logistics policy alternatives
Christopher [14]	Logistics and competitive advantage	Supply chain and logistics management	Connects customer value with total logistics cost and service differentiation	Does not explicitly evaluate competing e-commerce delivery and return policies
Christopher and Peck [15]	Integration of marketing and logistics	Cross-functional value delivery	Demonstrates the need to coordinate demand generation with fulfillment capability	Provides a conceptual rather than multi-criteria policy-ranking framework
Kawa and Zdrenka [8]	Logistics value, satisfaction, loyalty, and retailer performance	E-commerce logistics service	Shows the relationship between logistics value and customer/business outcomes	Does not jointly evaluate delivery incentives, return risks, and financial contribution as policy alternatives
Akul and Ungan [16]	E-commerce logistics service quality	Customer satisfaction and loyalty	Identifies timeliness, order condition, accuracy, and discrepancy handling as important service dimensions	Focuses primarily on service quality rather than structured policy selection
DHL [6, 7, 17]	Delivery preferences, return options, trust, and out-of-home delivery	Industry and consumer behavior	Provides evidence that delivery and return conditions affect purchase completion and customer expectations	Does not provide a formal analytical model for multi-criteria policy evaluation
Shopify [19, 20]	Return-policy design and management	Retail practice and customer communication	Highlights transparency, accessibility, and operational management of returns	Focuses on policy practice rather than quantitative comparison of alternative logistics strategies

The distinction between created orders and realized sales is especially important. Marketing reports frequently focus on traffic, conversion, and the number of orders generated. Logistics operations, however, determine whether these orders are dispatched, successfully delivered, and ultimately retained after possible returns. Advertising expenditure should therefore be considered in relation to completed and retained order rate rather than only the number of orders initially created. Average order value should likewise be examined together with transportation costs, packaging costs, failed-delivery risks, and return-related expenditure. Without this integrated perspective, an apparently successful policy may generate weaker economic performance once fulfillment outcomes are incorporated.

Delivery and return policies also need to be evaluated in relation to differences among customer segments and purchase contexts. Customers purchasing low-value products may be particularly sensitive to delivery charges because transportation represents a relatively large proportion of total expenditure. Customers purchasing gifts during holiday periods may place greater emphasis on delivery reliability. Other customers may value parcel-locker delivery, shipment redirection, advance-payment incentives, or simplified returns. A single policy is therefore unlikely to perform equally well across all customer and operational conditions. This reinforces the need for a structured evaluation approach capable of accommodating multiple and potentially conflicting performance criteria.

The same reasoning applies to logistics differentiation. Retailers can compete through faster or more reliable delivery, clearer communication, appropriate packaging, flexible receiving options, and simpler return procedures. Such improvements do not necessarily require the most expensive logistics configuration; in some cases, better information and more consistent fulfillment can create substantial customer value. However, each service improvement carries operational requirements and potential costs. Consequently, the value of a logistics policy should be assessed through its overall consequences rather than through a single customer-service or cost indicator.

A further issue concerns the use of logistics information as feedback for marketing and operational improvement. Return reasons can indicate weaknesses in product descriptions, photographs, packaging, order accuracy, supplier quality, or delivery performance. Unclaimed shipments can reveal problems associated with particular payment methods, customer segments, products, or promotional campaigns. These data therefore provide information not only about reverse logistics but also about the quality of upstream marketing and fulfillment decisions. Linking such indicators within a common evaluation structure can support a more realistic assessment of alternative policies.

Although previous studies have extensively examined delivery services, return management, customer satisfaction, and the financial implications of e-commerce logistics, their findings remain largely fragmented across individual dimensions. A significant research gap therefore persists in the integrated evaluation of competing delivery and return policies, particularly regarding the simultaneous consideration of customer-related outcomes, operational performance, logistics risks, and financial sustainability within a structured multi-criteria decision-making framework.

3 Distribution Logistics and Reverse Logistics Processes

The e-commerce distribution process consists of a sequence of interdependent activities extending from order receipt and verification to final delivery to the customer. Its performance depends not only on the speed of individual operations but also on the accuracy of information, inventory availability, order picking, packaging, courier handover, and last-mile execution. Because these activities are closely connected, errors occurring at an early stage can propagate through the subsequent fulfillment process and generate additional costs, unsuccessful deliveries, product returns, or customer dissatisfaction.

The process begins with order receipt and verification. Before an order is processed, product information, inventory records, delivery charges, and payment options must be correctly configured within the online store. Once an order has been submitted, customer information, particularly the delivery address, telephone number, city, and postal code, should be checked for completeness and consistency. Automated validation of mandatory fields can reduce routine manual intervention and enable employees to concentrate on potentially problematic orders. Product availability should also be confirmed at this stage, and the required quantities should be reserved. This control becomes particularly important during periods of increased demand, when discrepancies between online inventory records and physical stock can result in cancellations, delayed fulfillment, and customer dissatisfaction.

Order picking, verification, packaging, and preparation for courier handover form the next stage of the outbound process. Before a parcel is closed, the selected products and quantities should be checked against the order record. Packaging should reflect the characteristics of the product, including its size, weight, and fragility. Standardized packaging procedures can reduce preparation time, limit unnecessary material consumption, and lower the probability of damage during transportation. Packaging therefore represents both a direct logistics cost and an operational factor influencing delivery quality, customer satisfaction, and the likelihood of subsequent complaints or returns. After verification and packaging, the parcel is labeled and transferred to the courier, while relevant shipment information, including the tracking number and cash-on-delivery amount where applicable, is recorded in the system.

Last-mile delivery represents the final stage of outbound logistics and is particularly important because it is the point at which customers directly experience the quality of fulfillment. Delivery performance should therefore be evaluated through multiple indicators rather than delivery speed alone. Relevant measures include delivery time, first-attempt delivery success, shipment damage, tracking accuracy, complaint resolution, and the reliability of courier status information. During periods of increased demand, such as holiday seasons, limited courier capacity can extend lead times and increase the probability of unsuccessful delivery attempts. Under these conditions, delivery commitments should be communicated realistically, while shipment tracking and flexible delivery or pickup options can help reduce avoidable failures.

Unsuccessful and unclaimed shipments constitute a distinct source of operational and financial risk in e-commerce logistics. Failure to complete delivery may result from incorrect customer information, customer unavailability, refusal to accept the parcel, errors in the cash-on-delivery amount, or courier-related problems. When a shipment is returned to the retailer, the resulting loss extends beyond the reverse transportation charge. Additional costs may arise from receiving and inspecting the parcel, repackaging, updating inventory records, internal processing, and the advertising expenditure associated with acquiring the original order. The product may also remain temporarily unavailable for resale while it is in transit or undergoing inspection and processing. Consequently, the unclaimed shipments should be treated as both a logistics-performance indicator and a financial-risk measure, particularly in business models where cash on delivery (COD) remains common.

Reverse logistics begins when products move from customers back to the retailer, supplier, service provider, or another location where they can be inspected and processed. Returns may arise because the delivered product does not correspond to customer expectations, because of retailer errors, transportation damage, delivery delays, product defects, or a customer decision to withdraw from the purchase. Compared with outbound distribution, reverse flows are less predictable in both timing and product condition, which increases the importance of systematic recording and standardized processing procedures.

A structured return process should begin with a clearly defined request procedure in which the order number, reason for return, and relevant supporting information are recorded. Once the returned product is received, it should be inspected to determine whether it is complete, undamaged, and suitable for resale. Depending on its condition, the item may be returned directly to inventory, repackaged and sold at a reduced value, repaired or returned to the supplier, or written off. The total cost of a return may therefore include transportation, communication, handling, inspection, repackaging, refund processing, potential loss of product value, and the marketing expenditure associated with the original sale.

For this reason, return performance should not be assessed solely through the number of products sent back by customers. Measures such as the return rate, average return cost, expected return cost per delivered order, and the distribution of return reasons provide a more informative basis for evaluating the operational and financial consequences of reverse logistics. Incorporating these measures into profitability analysis is particularly important because successfully delivered orders do not necessarily remain realized sales once subsequent returns are considered.

Return information also provides feedback for improving the outbound logistics process. Return reasons should therefore be recorded using consistent categories, such as expectation mismatch, retailer error, transportation damage, delivery delay, customer change of mind, and product defect. Repeated occurrences within a particular category may reveal problems in product descriptions, order picking, packaging, supplier quality, or courier performance. In this way, reverse logistics functions not only as a cost-generating activity but also as an information source for correcting weaknesses in the broader e-commerce fulfillment system.

Taken together, distribution logistics and reverse logistics form a continuous decision-relevant process extending from order creation to the final retention of a sale. The number of orders generated therefore provides only a partial representation of business performance. A more meaningful assessment distinguishes between created orders, dispatched shipments, successfully delivered orders, unclaimed shipments, product returns, and retained order rate. This distinction is central to the subsequent analysis because delivery and return policies can improve one stage of the process while weakening another. Evaluating such policies therefore requires the simultaneous consideration of customer response, fulfillment performance, logistics risk, and economic outcomes.

4 Practical Analysis of the Impact of Delivery and Returns on Sales

The practical analysis covers the period from November 15, 2025, to January 15, 2026, encompassing Black Friday, the New Year holiday season, and the first half of January (Table 2). This interval was selected because periods of intensified demand provide a particularly informative setting in which to examine the interaction between customer response and logistics performance. Website traffic and order volumes tend to increase, fulfillment operations face greater workloads, courier capacity becomes more constrained, and customers may become more sensitive to delivery reliability and timing. At the same time, promotional activity may stimulate impulse purchases, increasing the probability that some orders will subsequently remain unclaimed or be returned. The period therefore provides an appropriate analytical setting for examining how delivery and return policies affect both realized sales and logistics-related costs.

Table 2. Overview of modeled business performance indicators during the period of increased sales

Period	Website Visits	Orders	Dispatched Shipments	Delivered Shipments	Unclaimed Shipments	Returns	Average Order Value (RSD)
November 15–30, 2025	23,500	420	402	358	44	16	2,280
December 1–15, 2025	24,000	510	490	442	48	20	2,350
December 16–31, 2025	31,500	740	705	632	73	31	2,980
January 1–15, 2026	22,000	390	371	326	45	18	2,420
Total	101,000	2,060	1,968	1,758	210	85	–

Note: "—" indicates that the value is not applicable to the total row; RSD = Serbian dinar.

The modeled retailer sells small household products, automotive accessories, lighting products, and basic electronic devices. COD represents the dominant payment method, while a smaller proportion of orders are prepaid. Monetary values are expressed in Serbian dinars (RSD), the official currency of Serbia. To facilitate international interpretation, the official middle exchange rate published by the National Bank of Serbia on January 15, 2026, was USD 1 = RSD 100.8096 [21]. The RSD is retained as the calculation currency throughout the analysis to preserve consistency across the modeled cost and revenue structure. Thus, the standard delivery charge of RSD 400 corresponds to approximately USD 3.97, the average courier cost of RSD 420 to approximately USD 4.17, and the RSD 3,000 free-delivery threshold to approximately USD 29.76.

The standard customer delivery charge was RSD 400, whereas the contracted average courier cost was estimated at RSD 420 per outbound shipment. Average packaging expenditure was estimated at RSD 72 per shipment. Three principal delivery conditions were represented in the modeled study period: (1) customer-paid delivery, (2) unconditional free delivery, and (3) free delivery for orders exceeding RSD 3,000. These conditions provide the operational basis for examining whether policies designed to stimulate purchasing activity generate sufficient additional value to compensate for the corresponding logistics expenditure.

The analysis incorporates website visits, orders created, dispatched shipments, successfully delivered shipments, unclaimed shipments, product returns, average order value, and promotional expenditure. In addition to these conventional indicators, a retained-sales measure was used. Retained orders are defined as successfully delivered orders minus subsequent product returns. This distinction is important because an order recorded in the online store does not necessarily become a completed and economically retained transaction.

During the analyzed period, 2,060 orders were created, of which 1,968 were dispatched. The remaining 92 orders were associated with pre-dispatch cancellations, unresolved customer-data errors, or unavailable products. Of the dispatched shipments, 1,758 were successfully delivered and 210 were returned as unclaimed. A further 85 orders were returned after successful delivery, producing 1,673 retained purchases. These differences illustrate why order creation alone provides an incomplete representation of realized business performance.

Delivery cost also depends on who bears the transportation charge and whether the transaction is ultimately retained. When the customer pays RSD 400 for delivery and the retailer pays the courier RSD 420, the retailer bears a net delivery cost of RSD 20 for a successfully completed transaction. Under unconditional free delivery, the retailer bears the entire RSD 420 courier charge. If a shipment is unclaimed or subsequently returned, additional reverse-logistics expenditure is incurred.

In addition to transportation costs, each dispatched order incurs an average packaging cost of RSD 72. The model also incorporates an average payment-processing cost of RSD 35 per retained sale and the promotional expenditure incurred during the analyzed period. Product acquisition cost was assumed to equal approximately 43% of retained-sales revenue. This parameter forms part of the modeled business structure and was selected to represent a retail configuration in which sufficient gross margin remains to cover marketing, logistics, and other variable operating costs.

For an unclaimed shipment, the retailer receives neither product revenue nor delivery revenue, even though packaging, outbound transportation, and customer-acquisition costs have already been incurred. Because outbound courier and packaging expenditure is incorporated for all dispatched parcels, an additional RSD 270 was assigned to the return transportation and internal handling of each unclaimed shipment. For products returned after successful delivery, an additional cost of RSD 570 was assumed, covering return transportation, inspection, repackaging, and the estimated average loss in product value.

The economic difference between delivery policies can be illustrated using an order valued at RSD 2,400. Under customer-paid delivery, the retailer bears a net courier cost of only RSD 20. Under free delivery, this rises to RSD 420. The RSD 400 difference represents a substantial proportion of the available contribution when product margins are moderate. This does not imply that free delivery is necessarily economically unfavorable. Such a policy may stimulate additional purchases or increase average order value. The relevant decision question is therefore whether these gains are sufficient to offset the additional logistics expenditure.

The modeled period from December 1 to December 15 represents unconditional free delivery. Although the number of orders and purchase conversion increased relative to the preceding period, the resulting contribution after major variable costs was substantially weaker. During the second half of December, free delivery was instead linked to a minimum order value of RSD 3,000. Average order value increased to RSD 2,980. Within the logic of the modeled scenario, the threshold is assumed to encourage some customers whose baskets are close to the eligibility level to add an additional product.

Delivery expenditure should therefore be evaluated relative to order value. A courier cost of RSD 420 represents 17.5% of an order valued at RSD 2,400 but approximately 13.8% of an order valued at RSD 3,050. As basket value increases, the same transportation charge accounts for a smaller proportion of the transaction value and may be more readily absorbed by the available product margin. A free-delivery threshold is consequently economically defensible only when the incremental contribution generated by additional purchases is sufficient to compensate for the delivery cost.

Table 3. Example of the cost of a single order under three delivery models

Item	Customer-Paid Delivery	Free Delivery for All Orders	Free Delivery Above RSD 3,000
Product value	2,400	2,400	3,050
Purchase cost	1,050	1,050	1,330
Packaging	72	72	78
Payment processing	35	35	40
Marketing cost per order	520	520	560
Net delivery cost	20	420	420
Expected logistics risk	116	116	122
Contribution per order (RSD)	587	187	500

Note: RSD = Serbian dinar.

The economically appropriate threshold is retailer-specific. It depends on average product margins, courier charges, packaging costs, customer-acquisition expenditure, and the expected incidence of unclaimed shipments and subsequent returns. Expected logistics risk in the present model represents the estimated average cost associated

with these adverse fulfillment outcomes, with a slightly higher amount assigned to higher-value orders. Table 3 illustrates the resulting cost structure under three delivery conditions.

Contribution per order was calculated as:

Contribution per order = product and delivery revenue - purchase cost - packaging - payment processing - marketing cost - delivery cost - expected logistics risk.

4.1 Analysis of Unclaimed Shipments

The unclaimed shipment rate was calculated as the number of unclaimed shipments divided by the total number of dispatched shipments. Of the 1,968 parcels dispatched during the analyzed period, 210 were not collected, producing an overall unclaimed shipments of 10.67%. In practical terms, approximately one in every ten dispatched parcels failed to result in a completed delivery.

The highest rate was observed during the first half of January, at 12.13%. Within the modeled setting, the post-holiday period may be associated with weaker purchasing commitment following seasonal expenditure, while some purchases initiated during promotional periods may have been more impulsive. Non-working days and delivery delays may also contribute to customers reconsidering purchases before receipt. The corresponding rates were 10.95% during the second half of November, 9.80% during the first half of December, and 10.35% during the second half of December. The fact that the period with the highest shipment volume did not exhibit the highest unclaimed rate indicates that volume alone does not determine unsuccessful-delivery performance.

The modeled total cost of an unclaimed shipment was RSD 760, comprising RSD 420 for outbound transportation, RSD 220 for return transportation, RSD 70 for packaging, and RSD 50 for internal handling. For 210 unclaimed shipments, the resulting direct logistics cost was RSD 159,600. Advertising expenditure is not included in this amount and therefore represents an additional economic exposure. When the direct unclaimed-shipment cost is distributed across all 1,968 dispatched parcels, it corresponds to approximately RSD 81.10 per shipment. This illustrates how unsuccessful deliveries create a risk cost that must ultimately be absorbed by successful transactions.

Payment method shows an important difference within the modeled dataset. Of 1,770 cash-on-delivery shipments, 204 were unclaimed, corresponding to 11.53%. Among 198 prepaid shipments, six unsuccessful deliveries occurred, producing a rate of 3.03% (Table 4). The modeled comparison therefore associates prepayment with a substantially lower probability of an unclaimed shipment.

Table 4. Unclaimed shipments by payment method

Payment Method	Dispatched	Unclaimed	Unclaimed Shipments
Cash on delivery (COD)	1,770	204	11.53%
Prepayment	198	6	3.03%
Total	1,968	210	10.67%

This result should not be interpreted as evidence that COD should be eliminated. For customers purchasing from an unfamiliar retailer, the option may reduce perceived transaction risk and support trust. A more balanced policy is to encourage prepayment while retaining COD for customers who prefer it. Possible incentives include modest delivery discounts, priority processing, or benefits attached to subsequent purchases. These characteristics provide the operational rationale for including a prepayment incentive as a distinct policy component in the subsequent multi-criteria evaluation.

Order confirmation provides an additional risk-control mechanism. In the modeled operating process, automated confirmation enables customers to verify an order or correct address information before dispatch. Manual telephone verification can then be reserved for cases involving incomplete information, unusually large quantities, or previous unclaimed shipments. Such selective intervention concentrates administrative effort on orders for which verification is most likely to reduce fulfillment risk.

Unclaimed shipments should also be monitored by traffic source, product category, and promotional campaign. A campaign generating a large number of low-commitment orders may perform less favorably once unsuccessful deliveries are incorporated into the assessment. Linking online-store order records with courier status information therefore enables the retailer to distinguish nominal customer acquisition from successfully completed transactions.

The overall rate and associated cost were calculated as follows:

Unclaimed shipments = (210 ÷ 1,968) × 100 = 10.67%

Total direct cost of unclaimed shipments = 210 × RSD 760 = RSD 159,600.

4.2 Analysis of the Product Return Rate

The product return rate was calculated relative to successfully delivered shipments because a post-delivery return can occur only after the customer has received the order. Of the 1,758 successfully delivered orders, 85 were

subsequently returned, corresponding to a return rate of 4.84%. The lowest modeled rate occurred during November, at 4.47%, whereas the highest occurred during the first half of January, at 5.52%.

The January pattern is consistent with a post-holiday operating context in which customers have more time to evaluate purchases and some products may have been acquired as gifts. Delayed holiday deliveries may also lose their intended purpose even when the product itself is technically satisfactory. These contextual factors should be interpreted cautiously because the analysis is scenario-based rather than a controlled causal experiment.

The additional cost assigned to each post-delivery return was RSD 570. This amount consists of RSD 360 for return transportation, RSD 90 for receiving and repackaging, and RSD 120 for the estimated average reduction in product value. For 85 returns, the resulting additional cost was RSD 48,450. The full acquisition cost of returned products was not treated as lost because part of the merchandise can be restored to inventory and resold.

Allocated across the 1,758 successfully delivered orders, the additional return cost corresponds to approximately RSD 27.56 per delivery. When considered together with the cost generated by unclaimed shipments, logistics risk exceeds RSD 100 per dispatched order. This amount is sufficiently material to warrant explicit consideration in pricing, promotional planning, and delivery-policy evaluation.

The distribution of return reasons provides more decision-relevant information than the overall return rate alone. The largest category consisted of products that did not meet customer expectations or did not sufficiently correspond to their online description, representing 29 cases or 34.1% of all returns (Table 5). This category points to potential improvements in product photography, dimensional information, and product descriptions. Although the financial consequence materializes through reverse logistics, the underlying source of the problem may originate in product presentation.

Table 5. Structure of return reasons in the modeled scenario

Reason	Number	Share (%)
Product did not meet expectations or match the description	29	34.1
Damage during transportation	16	18.8
Customer changed their mind	15	17.6
Delivery delay	13	15.3
Incorrect or incomplete item	8	9.4
Product defect	4	4.7
Total	85	100.0

Transportation damage accounted for 18.8% of returns and indicates a need to examine packaging standards and courier handling. Customer reconsideration represented 17.6%, delivery delays 15.3%, incorrect or incomplete items 9.4%, and product defects 4.7%. These categories require different managerial responses. Incorrect or incomplete items and product defects together represented slightly more than 14% of returns and are comparatively controllable through order verification, supplier quality control, and pre-dispatch inspection.

Delivery delays illustrate the interaction between logistics performance and the economic value of an order. A product may remain technically satisfactory while no longer serving the purpose for which it was purchased. Holiday delivery commitments should therefore be adjusted before fulfillment capacity becomes insufficient to meet them. The short-term revenue sacrificed by imposing an earlier delivery cutoff may be lower than the combined cost of returns, customer dissatisfaction, and after-sales processing.

Return reasons should consequently be interpreted as indicators of marketing, procurement, fulfillment, and logistics quality rather than merely as reverse-logistics classifications. A more restrictive return policy would not necessarily resolve problems caused by inaccurate product information, packaging deficiencies, fulfillment errors, or delivery delays. Accordingly, return-management improvements should be evaluated as one component of a broader logistics policy rather than as an isolated basis for policy selection.

The return rate and corresponding additional cost were calculated as follows:

Return rate = (85 ÷ 1,758) × 100 = 4.84%

Total additional return cost = 85 × RSD 570 = RSD 48,450.

4.3 Impact of Free Delivery on the Number of Orders

The relationship between delivery policy and purchasing activity was examined across four consecutive observation periods (Table 6). Customer-paid delivery was applied during the first and fourth periods, unconditional free delivery during the second, and free delivery above RSD 3,000 during the third.

Because these periods differed in seasonality, advertising intensity, and underlying demand, the comparison does not establish that delivery policy alone caused the observed changes. It should instead be interpreted as an

exploratory scenario analysis demonstrating how delivery conditions interact with conversion, order value, and logistics expenditure.

Table 6. Delivery policy and sales indicators by period

Period	Policy	Purchase Conversion Rate	Average Order Value (RSD)	Advertising Cost per Order (RSD, Rounded)
November 15–30	Customer pays RSD 400	1.79%	2,280	512
December 1–15	Free delivery for all orders	2.13%	2,350	510
December 16–31	Free delivery above RSD 3,000	2.35%	2,980	568
January 1–15	Customer pays RSD 400	1.77%	2,420	526

Note: RSD = Serbian dinar.

During November 15–30, when customers paid the RSD 400 delivery fee, the purchase conversion rate was 1.79%. During December 1–15, unconditional free delivery coincided with an increase in conversion to 2.13% and an increase in orders from 420 to 510 despite similar website traffic. This pattern is consistent with the possibility that removal of the delivery charge supported purchase completion. However, average order value increased only from RSD 2,280 to RSD 2,350. Because the retailer absorbed the RSD 420 courier charge for each dispatched parcel, the higher number of transactions did not translate into a stronger contribution after major variable costs.

During December 16–31, the free-delivery threshold was set at RSD 3,000. Purchase conversion reached 2.35%, while average order value increased to RSD 2,980. The period also coincided with stronger seasonal demand and greater promotional activity; consequently, the observed improvement cannot be attributed exclusively to the threshold. Within the modeled decision setting, however, the threshold represents a mechanism through which customers with basket values close to RSD 3,000 may be encouraged to add another item.

When customer-paid delivery was reintroduced during January 1–15, the conversion rate decreased to 1.77% and average order value declined to RSD 2,420. The simultaneous end of the holiday season makes a causal interpretation inappropriate. A rigorous estimate of the isolated effect of delivery policy would require a controlled comparison involving similar customer groups, product availability, traffic quality, and advertising expenditure.

The comparison nevertheless illustrates the central trade-off examined in this study. Unconditional free delivery can increase customer attractiveness while transferring transportation expenditure to the retailer. A threshold-based policy can potentially increase basket value while limiting the proportion of low-value orders for which the retailer absorbs the delivery charge. Customer-paid delivery preserves a larger contribution per transaction and limits direct logistics-cost exposure, although it may discourage purchases when the shipping charge represents a substantial proportion of the order value. The relative attractiveness of these policies therefore depends on the weights assigned to customer response, financial contribution, logistics risk, and implementation requirements.

The appropriate threshold should therefore be determined relative to the existing order-value distribution. If average order value is approximately RSD 2,300, a threshold of RSD 10,000 is unlikely to influence purchasing behavior. Conversely, a threshold of RSD 2,500 may be so close to existing spending levels that many customers qualify without modifying their baskets. A threshold around RSD 3,000 is sufficiently close to the modeled average order value to create a plausible incentive for an incremental purchase while still requiring a meaningful increase for many customers.

A free-delivery incentive may also be combined with prepayment. Such a policy can simultaneously influence basket size and reduce exposure to unclaimed cash-on-delivery orders. The condition should, however, be communicated transparently so that customers do not interpret it as a penalty for selecting COD.

From an analytical perspective, promotional delivery policies should be specified in advance and evaluated through several indicators simultaneously, including purchase conversion, average order value, contribution, unclaimed shipments, and post-delivery returns.

4.4 Comparison of Paid and Free Delivery Scenarios

To reduce the influence of differences in website traffic, an additional normalized scenario based on 10,000 website visits was developed (Table 7). The product assortment and traffic quality were assumed to remain constant, while delivery conditions, purchase conversion, and average order value differed across the three scenarios. The purpose of this comparison is not to estimate causal effects but to illustrate the economic consequences of alternative delivery structures under comparable traffic conditions.

Table 7. Comparison of three scenarios based on 10,000 website visits

Scenario	Purchase Conversion Rate	Number of Orders	Contribution Per Order (RSD)	Total Contribution (RSD)
Customer pays for delivery	1.80%	180	587	105,660
Free delivery for all orders	2.20%	220	187	41,140
Free delivery above RSD 3,000	2.15%	215	500	107,500

Note: RSD = Serbian dinar.

Under customer-paid delivery, a conversion rate of 1.80% was assumed, corresponding to 180 orders. With a contribution of RSD 587 per order, total contribution reached RSD 105,660. This configuration produced the lowest number of purchases but the highest contribution per order.

Under unconditional free delivery, conversion increased to 2.20%, producing 220 orders. Contribution per order, however, fell to RSD 187, resulting in total contribution of only RSD 41,140. The comparison illustrates how higher order volume can create an apparently favorable commercial result while producing a substantially weaker economic outcome after delivery expenditure is incorporated.

For free delivery above RSD 3,000, a conversion rate of 2.15% and 215 orders were assumed. Contribution per order amounted to RSD 500, producing total contribution of RSD 107,500. Under the modeled assumptions, this scenario combined a greater number of purchases than customer-paid delivery with slightly higher total contribution.

This result should not be generalized to all e-commerce environments. A threshold is advantageous only when the additional basket value generates sufficient margin to compensate for the delivery cost. If incremental products have very low margins, the higher order value may provide little economic gain. Similarly, if the pre-existing average order value already exceeds the threshold, the policy may provide free delivery without materially changing customer purchasing behavior. The threshold should therefore be calibrated according to the retailer's product mix, margins, and order-value distribution.

The scenario also does not incorporate longer-term effects on customer retention. Free delivery may improve the customer experience and potentially strengthen repeat purchasing. Conversely, a financially unsustainable delivery policy may reduce the resources available for packaging quality, customer support, or other service dimensions. Short- and longer-term consequences should therefore be considered jointly.

A more robust empirical evaluation would require a controlled test over comparable periods, ideally with similar products, customer segments, traffic sources, and promotional budgets. Such testing should evaluate retained order rate, average order value, contribution, unclaimed shipments, and returns in addition to created orders. The broader implication is that a delivery policy is simultaneously a customer, logistics, and financial decision: its attractiveness before purchase must ultimately be evaluated against the costs and risks that emerge through fulfillment. The scenario comparison in Table 8 focuses on contribution under standardized traffic assumptions and should not be interpreted as a complete ranking of the delivery policies. The subsequent multi-criteria analysis additionally incorporates retained-order performance, unsuccessful-delivery risk, return exposure, logistics cost, and implementation complexity, which may alter the overall preference ordering.

Table 8. Proposed daily overview of operational indicators

Indicator	Measurement Method	Decision Relevance
Orders received by 12:00	Number of orders recorded in the administrative system	Daily workload planning
Orders dispatched on the same day	Number dispatched / number ready for dispatch	Fulfillment-speed monitoring
Data errors	Number of orders requiring customer contact	Assessment of order-form and traffic-source quality
Incorrectly packed orders	Number of confirmed packing errors	Order-picking control
Failed first delivery attempt	Courier-status records	Early identification of unsuccessful-delivery risk
Returns received and resolved	Number of returns and average processing time	After-sales service performance

4.5 Description of the Online Store Business Model

The analyzed online retailer operates in the domestic market, offering household products, lighting equipment, small household appliances, automotive accessories, and basic electronic devices. The product assortment predominantly

belongs to the low- and medium-price segments, with best-selling items priced between RSD 1,490 and RSD 3,990. Given the relatively modest transaction values, delivery expenses represent a significant component of the total purchase cost and can substantially influence customer purchasing decisions.

Sales are conducted through the retailer's own e-commerce platform, with paid social-media advertising representing an important source of website traffic and customer acquisition. Inventory is maintained in a small warehouse, while transportation services are outsourced to a contracted courier company. COD is the prevailing payment method, although customers can also select prepayment. Under standard conditions, the delivery charge amounts to RSD 400 per order, while promotional arrangements include free delivery and conditional free delivery for orders exceeding RSD 3,000. The latter policy may encourage customers to increase their basket value by adding complementary products to reach the required threshold.

The financial performance of this business model is particularly sensitive to customer-acquisition expenses, transportation charges, unsuccessful deliveries, and product returns. Consequently, the difference between the selling price and product acquisition cost does not adequately represent the actual contribution generated by an order. Advertising, packaging, payment processing, outbound transportation, and costs associated with unclaimed shipments and returns must also be considered when evaluating the economic sustainability of alternative delivery policies.

The analyzed period encompasses the Black Friday and New Year shopping seasons, characterized by intensified promotional activities, potentially higher order volumes, and increased operational uncertainty. Holiday-related disruptions, courier capacity constraints, and impulse purchasing may influence both delivery reliability and the proportion of unclaimed shipments. These characteristics make the selected business model particularly suitable for examining the relationships between delivery incentives, customer purchasing behavior, logistics performance, and financial sustainability.

4.6 Order Receipt, Processing, and Dispatch Process

The retailer's order fulfillment activities are supported by an e-commerce information system that records customer details, ordered products, quantities, prices, discounts, delivery charges, and selected payment methods. Given the limited operational resources and the seasonal variability of demand, customer-data verification is performed selectively rather than through systematic manual confirmation of every order. Particular attention is given to orders containing incomplete contact information, inconsistent delivery addresses, or other indications of potential fulfillment problems. Inventory availability is additionally verified against physical stock before dispatch to reduce the risk of unsuccessful order processing.

Warehouse operations are organized around a relatively small inventory and limited staffing capacity. Frequently ordered products are positioned near the packing area, while additional accuracy checks are applied to multi-item orders. Packaging requirements vary according to product characteristics, particularly for fragile household and electronic items. The information system supports the preparation of shipping documentation and the transfer of COD amounts to the courier shipment record, reducing the need for manual data entry.

Under standard operating conditions, parcels are handed over to the contracted courier once per day. During the higher-demand period from 16 to 31 December, additional courier collections were introduced on selected days to reduce dispatch delays. Shipment status is monitored through courier tracking information, and customer contact is initiated when delivery exceptions occur, particularly in cases involving address inconsistencies or customer unavailability. These interventions are intended to reduce the proportion of unclaimed shipments and the associated transportation costs.

Return management is characterized by a relatively low level of digital integration. Customer communication takes place through email and telephone, while return-related information is maintained in spreadsheets rather than a dedicated returns-management application. The recorded information includes order identification, return reasons, relevant dates, product condition, associated costs, and final resolution. This arrangement offers a low-cost administrative solution but increases dependence on accurate manual data recording.

The described operational configuration provides the basis for monitoring fulfillment-related indicators, including order confirmation time, time to courier handover, same-day dispatch rate, successful delivery rate, order-processing errors, unclaimed shipment rate, damaged shipments, and return-processing time (Table 8). These indicators characterize the operational constraints and performance of the analyzed retailer and provide relevant information for assessing alternative delivery and return policies.

4.7 Delivery and Product Return Problems

The analysis identifies several interrelated sources of logistics and financial risk.

First, the high proportion of cash-on-delivery transactions creates exposure to unclaimed shipments. The modeled unclaimed rate of 11.53% for cash-on-delivery shipments indicates that part of the logistics effort and customer-acquisition expenditure does not result in revenue. The corresponding rate is substantially lower for prepaid orders.

Second, expectation mismatch represents the largest category of product returns. When customers perceive that the received item differs from its online presentation, the financial consequence appears in reverse logistics even though the underlying problem may originate in marketing information. Product photographs, visible dimensions, usage descriptions, and specification clarity therefore form part of return-risk management.

Third, transportation damage reflects the interaction between packaging practice and courier performance. Periods of increased order volume can place pressure on packing operations and transportation networks. Standardized packaging procedures are therefore preferable to protection levels that vary according to employee workload.

Fourth, holiday delivery deadlines create a risk of expectation failure. Orders placed close to major holidays may not realistically arrive within normal lead times. Maintaining standard delivery promises under constrained courier capacity can therefore generate avoidable dissatisfaction and returns. In the modeled dataset, delivery delays accounted for 13 post-delivery returns.

Fifth, unconditional free delivery increases cost exposure. Although this policy coincided with higher order activity, the increase in average order value was insufficient to compensate for the transportation expenditure absorbed by the retailer. The modeled result illustrates why delivery incentives should be evaluated through contribution rather than order volume alone.

Sixth, inconsistent classification of return reasons reduces the analytical value of operational data. A generic courier status such as ”returned” does not distinguish between customer refusal, customer absence, address error, or another cause. Similarly, a description such as ”dissatisfied” provides insufficient information for selecting corrective action. More precise and standardized categories are therefore required.

Seventh, fragmented return handling can extend processing time. When part of a customer case remains in email or telephone communication while other information is recorded separately, both employees and customers may need to reconstruct the history of the case. A unique case identifier and standardized processing status can improve traceability.

These problems are not independent. Promotional campaigns can increase impulsive purchasing and fulfillment workload; higher warehouse pressure can increase packing errors; constrained courier capacity can increase delays; and these failures can subsequently increase returns and customer-service workload. The appropriate response is therefore not limited to a single logistics activity but requires coordination across marketing, sales, fulfillment, and reverse logistics.

4.8 Impact of Logistics Costs on Profitability

Financial performance was evaluated on the basis of retained order rate, defined as delivered orders remaining after post-delivery returns (Table 9). A total of 1,673 transactions were retained during the modeled period.

Table 9. Contribution by period after major variable costs

Period	Retained Orders	Product Revenue (RSD)	Delivery Revenue (RSD)	Contribution (RSD)	Margin
November 15–30	342	779,760	136,800	135,509	14.8%
December 1–15	422	991,700	0	25,059	2.5%
December 16–31	601	1,790,980	93,756	289,340	15.4%
January 1–15	308	745,360	123,200	127,333	14.7%
Total	1,673	4,307,800	353,756	577,241	12.4%

Note: RSD = Serbian dinar.

Product revenue from retained order rate amounted to RSD 4,307,800, while charged delivery fees generated RSD 353,756. Total modeled revenue therefore reached RSD 4,661,556. Major variable costs included product acquisition, estimated at 43% of retained product revenue; outbound courier charges for all 1,968 dispatched shipments; packaging; payment processing; advertising expenditure; additional costs associated with unclaimed shipments; and post-delivery return costs.

After these items were incorporated, total contribution amounted to RSD 577,241, representing approximately 12.38% of modeled revenue. This amount should not be interpreted as accounting profit because fixed expenses are excluded.

The weakest financial outcome occurred during December 1–15, when unconditional free delivery was applied. Although 422 retained orders generated product revenue of RSD 991,700, contribution was only RSD 25,059, corresponding to a margin of 2.5%. The principal modeled reason was that the retailer absorbed delivery costs without obtaining a sufficiently large increase in average order value.

The strongest absolute contribution occurred during December 16–31. During this period, 601 retained orders generated product revenue of RSD 1,790,980, and contribution reached RSD 289,340, representing a margin of 15.4%. Free delivery was restricted to orders exceeding RSD 3,000, while average order value reached RSD 2,980.

Under the model assumptions, this combination produced a more favorable balance between purchasing activity and transportation expenditure.

November and January generated similar margins of approximately 15%, largely because customers paid most of the delivery charge. January nevertheless exhibited higher unclaimed shipment and return rates, demonstrating that financial performance is not proportional to the number of orders alone.

Fixed expenses, including rent, owner remuneration, software, accounting, and taxation, were excluded. Consequently, the reported contribution represents the amount remaining after the principal variable costs and is available to cover fixed expenditure; it should not be interpreted as net profit.

The distinction between created and retained orders also affects the interpretation of marketing efficiency. Total advertising expenditure amounted to RSD 1,100,000. Dividing this amount by 1,673 retained transactions produces an advertising cost of approximately RSD 657.50 per retained sale. Dividing the same expenditure by the 2,060 orders initially created produces a substantially lower figure of RSD 533.98. The difference exceeds RSD 120 per transaction and demonstrates how marketing performance can be overstated when fulfillment outcomes are ignored.

A comparable difference is evident in revenue. The value of delivered transactions before post-delivery returns amounted to RSD 4,527,220, whereas retained product sales amounted to RSD 4,307,800. The difference of RSD 219,420 reflects sales value subsequently lost through returns or cancellations before the additional processing expenditure associated with these outcomes is considered.

The financial analysis therefore reinforces the central decision problem examined in this study. Delivery policy influences order creation and average basket value, while fulfillment performance determines how much of this nominal demand is converted into retained economic value. Logistics outcomes should consequently be incorporated into the assessment of marketing and delivery-policy performance rather than treated as a separate downstream concern. The financial results reported in this section represent only part of the overall policy evaluation; the subsequent multi-criteria analysis also incorporates customer response, logistics risk, return performance, and implementation complexity.

Advertising cost per retained sale = RSD 1,100,000 ÷ 1,673 = RSD 657.50

Advertising cost per created order = RSD 1,100,000 ÷ 2,060 = RSD 533.98.

4.9 Recommendations for Improving Logistics Service

The findings support several operational improvements that can also inform the alternative policies evaluated in the subsequent decision model.

First, customer-paid delivery should be retained as the baseline policy when contribution per order, logistics cost, and implementation simplicity are the dominant decision priorities. Threshold-based free-delivery policies may nevertheless provide a viable alternative when the objective is to improve customer attractiveness without assuming the full cost exposure associated with unconditional free delivery. The threshold should be periodically reassessed against changes in average order value, product margins, courier charges, and customer purchasing behavior. For lower-value products, bundled offers or complementary product combinations can provide an alternative mechanism for increasing basket value.

Second, prepayment should be encouraged without necessarily eliminating COD. Potential incentives include reduced shipping charges, priority processing, or a small voucher applicable to a future purchase. The purpose is to reduce exposure to unclaimed shipments while preserving payment flexibility.

Third, order verification should be risk-based. Automated checks can identify incomplete addresses, unusual quantities, previous unclaimed shipments, or other indicators requiring additional confirmation. Telephone verification can then be restricted to cases in which its expected value justifies the administrative cost.

Fourth, product information should be improved where return records indicate expectation mismatch. Best-selling products should provide sufficiently clear photographs, dimensions, examples of use, and limitations. More accurate pre-purchase information can reduce transactions that would otherwise appear as sales before subsequently becoming returns.

Fifth, packaging requirements should be standardized by product category. Minimum protection standards should remain consistent during peak periods rather than being relaxed when order volumes increase. Packaging for frequently ordered products may be prepared in advance where operationally appropriate.

Sixth, holiday delivery deadlines should be communicated conservatively and transparently. Websites and advertisements should specify the latest date for which pre-holiday delivery can reasonably be expected and should acknowledge the possibility of delays during periods of network congestion.

Seventh, reasons for unclaimed shipments and post-delivery returns should be recorded using standardized categories. Products, campaigns, customer-acquisition sources, and courier statuses associated with unusually high failure rates can then be identified through regular review.

Eighth, courier performance should be evaluated on multiple dimensions rather than price alone. Relevant indicators include delivery lead time, first-attempt delivery success, damage rate, status-information accuracy, and

complaint-resolution time. Where more than one courier is used, comparable performance data can support provider selection.

Ninth, the return procedure should be concise, visible, and consistently administered. Customers should receive clear instructions and confirmation that their request has been received, while employees should follow a standardized internal workflow.

Finally, marketing and logistics performance should be reviewed jointly. Before major promotional activity is launched, inventory availability, packing capacity, daily processing capability, and expected courier lead times should be assessed. After completion, campaign performance should be evaluated using retained orders, fulfillment outcomes, and contribution rather than created orders alone.

Taken together, these recommendations illustrate why delivery and return management constitutes a multi-criteria decision problem rather than a set of isolated operational choices. Policies that perform favorably from the perspective of customer acquisition may perform less favorably once logistics risk and economic contribution are incorporated. The modeled patterns and analytical relationships established in this section therefore provide the basis for defining the policy alternatives and evaluation criteria used in the decision framework developed in Section 5.

5 Development of the Decision Model for Evaluating Delivery and Return Policies

This study develops a structured decision-analytics framework for comparing alternative delivery and return policies in sustainable e-commerce logistics. The framework translates the operational and financial relationships identified in the preceding sections into a formal multi-criteria decision problem. It consists of four principal stages: (1) definition of the policy alternatives and evaluation criteria; (2) construction of the initial decision matrix; (3) objective determination of criterion weights using the MEREC method; and (4) ranking of the policy alternatives using the weighted MOOSRA method.

The framework is designed to capture the competing dimensions involved in delivery- and return-policy decisions. These dimensions include customer response, average order value, contribution per order, retained order rate, unsuccessful-delivery risk, post-delivery returns, logistics expenditure, and implementation complexity. By integrating benefit- and cost-oriented criteria within a common analytical structure, the model enables policies with different strengths and weaknesses to be compared on a consistent basis. The overall methodological structure is presented in Figure 1.

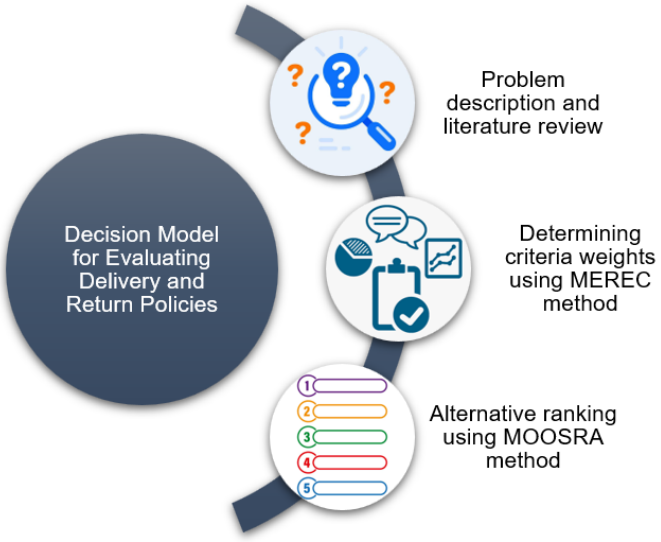


Figure 1. Proposed decision-analytics framework for evaluating delivery and return policies

Note: MEREC = Method based on the Removal Effects of Criteria; MOOSRA = Multi-Objective Optimization on the basis of Simple Ratio Analysis.

5.1 Method Based on the Removal Effects of Criteria Method

The MEREC was applied to derive objective criterion weights. The method determines the relative importance of each criterion according to the extent to which removal of that criterion changes the overall assessment of the alternatives. The procedure follows the framework proposed by Keshavarz-Ghorabaee et al. [22] and subsequently applied by Puška et al. [23].

The computational procedure consists of the following steps.

Step 1. Construction of the initial decision matrix.

Let $X = [x_{ij}]$ denote a decision matrix containing n alternatives and m criteria, where x_{ij} represents the performance of alternative i with respect to criterion j.

Step 2. Normalization of the decision matrix.

Because the criteria differ in orientation, the original values are transformed according to whether they are benefit- or cost-oriented:

$$n_{ij}^{x} = \begin{cases} \frac{\min_k x_{kj}}{x_{ij}}, & j \in B, \\ \frac{x_{ij}}{\max_k x_{kj}}, & j \in C. \end{cases} \tag{1}$$

where, n_{ij}^{x} denotes the normalized value, and B and C denote the sets of benefit- and cost-oriented criteria, respectively.

Step 3. Calculation of the overall performance measure.

The overall performance of alternative i is calculated as:

$$S_i = \ln \left[1 + \frac{1}{m} \sum_{j=1}^{m} \left| \ln \left(n_{ij}^{x} \right) \right| \right] \tag{2}$$

Step 4. Calculation of performance after criterion removal.

To assess the contribution of criterion j, the performance measure is recalculated after that criterion is removed:

$$S'_{ij} = \ln \left[1 + \frac{1}{m} \sum_{\substack{k=1 \\ k \neq j}}^{m} \left| \ln \left(n_{ik}^{x} \right) \right| \right] \tag{3}$$

Step 5. Determination of the removal effect.

The effect of removing criterion j is quantified using:

$$E_j = \sum_{i=1}^{n} \left| S'_{ij} - S_i \right| \tag{4}$$

A larger E_j indicates that the criterion has a stronger influence on the overall evaluation structure.

Step 6. Calculation of objective criterion weights.

The final weight of criterion j is calculated as:

$$w_j = \frac{E_j}{\sum_{k=1}^{m} E_k} \tag{5}$$

The resulting weights satisfy:

$$\sum_{j=1}^{m} w_j = 1$$

Criteria whose removal produces greater changes in the overall performance measure therefore receive greater weights.

5.2 Multi-Objective Optimization on the Basis of Simple Ratio Analysis Method

The MOOSRA method was used to rank the policy alternatives. The procedure follows Ray [11] and Vasić et al. [24]. MOOSRA evaluates each alternative by comparing its aggregated performance on benefit criteria with its aggregated performance on cost criteria. In the present framework, the objective weights obtained from MEREC are incorporated into the MOOSRA procedure.

Step 1. Construction of the decision matrix.

The same decision matrix used for MEREC is employed:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix} \tag{6}$$

where, x_{ij} represents the performance value of alternative i with respect to criterion j, with $i = 1, 2, \ldots, n, j = 1, 2, \ldots, m$, n denotes the number of alternatives and m denotes the number of evaluation criteria.

Step 2. Vector normalization.

The original values are converted into dimensionless values using:

$$x_{ij}^{*} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{n} x_{ij}^{2}}} \tag{7}$$

where, x_{ij}^{*} denotes the normalized value of alternative i under criterion j.

Step 3. Calculation of the weighted MOOSRA performance ratio.

Let the first g criteria be benefit-oriented and the remaining $m - g$ criteria be cost-oriented. The MOOSRA performance value is then calculated as:

$$Y_i = \frac{\sum_{j=1}^{g} w_j x_{ij}^{*}}{\sum_{j=g+1}^{m} w_j x_{ij}^{*}} \tag{8}$$

A higher Y_i indicates stronger overall performance.

For reference, when all criteria are treated as equally important, the unweighted form becomes:

$$Y_i = \frac{\sum_{j=1}^{g} x_{ij}^{*}}{\sum_{j=g+1}^{m} x_{ij}^{*}} \tag{9}$$

Eq. (9) is included only as the conventional equal-weight form. In this study, the final rankings were obtained using Eq. (8) because the criterion weights were derived using MEREC.

Step 4. Ranking of alternatives.

The alternatives are ranked in descending order of Y_i. The highest value corresponds to the highest-ranked alternative.

5.3 Application of the Integrated Method Based on the Removal Effects of Criteria-Multi-Objective Optimization on the Basis of Simple Ratio Analysis Framework

Five policy alternatives were defined on the basis of the operational patterns, cost relationships, and managerial issues identified in Sections 3 and 4 (Table 10).

Table 10. Description of the evaluated policy alternatives

Code	Alternative	Description
A1	Customer-Paid Delivery	The customer pays RSD 400 for delivery, with no free-delivery threshold; this represents the baseline policy.
A2	Free Delivery For All Orders	Delivery is free for all orders regardless of order value.
A3	Free Delivery Above RSD 3,000	Delivery is free for orders whose value exceeds RSD 3,000.
A4	Free Delivery Above RSD 3,000 + Prepayment Incentive	Threshold-based free delivery is combined with incentives encouraging advance payment.
A5	Threshold Delivery + Enhanced Return Management	Threshold-based free delivery is combined with standardized packaging, transparent return procedures, and systematic recording of return reasons.

Note: A = Alternative; RSD = Serbian dinar.

Eight evaluation criteria were defined (Table 11). Their selection reflects the relationships identified in the literature review and the operational and financial evidence developed in the preceding sections [24–27].

The alternatives were assessed using a five-point ordinal scale. For benefit criteria C1–C4, a higher score indicates more favorable performance, with 1 representing very unfavorable performance and 5 representing very favorable performance. For cost criteria C5–C8, the scale reflects the magnitude of the corresponding cost, risk, or implementation burden, with 1 representing a very low level and 5 representing a very high level. This directional distinction ensures consistency with the subsequent MEREC normalization and MOOSRA aggregation procedures, in which benefit criteria are maximized and cost criteria are minimized.

Because the analytical application of this study is based on a modeled e-commerce setting, the initial decision matrix was constructed as a scenario-based analytical assessment rather than as a survey-based expert elicitation exercise. The scores were assigned by jointly considering three sources of information: (1) the operational and financial relationships quantified in Section 4; (2) the qualitative characteristics of the five policy alternatives; and (3)

the evaluation dimensions established from the literature [25–28]. The same five-point scale was applied consistently across all alternatives and criteria. In cases where the preceding analysis provided direct evidence, such as relative conversion performance, contribution, unclaimed-shipment exposure, return behavior, or implementation burden, the score reflected the corresponding relative position of the alternative. Where no direct numerical observation was available, the score represented the relative policy effect implied by the modeled operating conditions. This procedure was adopted to ensure internal consistency between the practical analysis and the subsequent multi-criteria model while avoiding unsupported claims of external empirical measurement.

Table 11. Evaluation criteria

Code	Criterion	Type	Measurement Aspect
C1	Purchase Conversion Rate	Benefit	Ability of the policy to stimulate completed order creation
C2	Average Order Value	Benefit	Effect of the policy on average basket value
C3	Contribution per Order	Benefit	Contribution remaining after the principal variable costs are deducted
C4	Retained Order Rate	Benefit	Proportion of orders remaining as realized sales after delivery and subsequent returns
C5	Unclaimed Shipments	Cost	Exposure to orders dispatched but not accepted by customers
C6	Return Rate	Cost	Proportion of successfully delivered orders subsequently returned
C7	Logistics Cost per Order	Cost	Delivery, packaging, return, and expected logistics-risk expenditure
C8	Implementation Complexity	Cost	Operational and managerial complexity required to implement and maintain the policy

Note: C = Criteria.

The ordinal scores represent relative positions within the five-policy comparison rather than externally calibrated performance levels. A score of 5 on a benefit criterion therefore represents the most favorable relative performance among the modeled alternatives, whereas a score of 5 on a cost criterion represents the greatest relative cost, risk, or implementation burden. Because these scores are scenario-based rather than externally calibrated measurements, their influence on the final ranking is examined subsequently through systematic perturbation and sensitivity analysis.

The resulting decision matrix is presented in Table 12.

Table 12. Initial decision matrix

Alternative	C1	C2	C3	C4	C5	C6	C7	C8
A1	3	3	5	4	3	2	1	1
A2	5	3	1	2	4	3	5	1
A3	4	5	4	4	3	2	2	2
A4	4	5	4	5	1	2	2	3
A5	4	5	4	5	3	1	1	4

Note: A = Alternative; C = Criteria.

The initial matrix was first processed using MEREC. Application of Eq. (1) produced the normalized matrix shown in Table 13.

Table 13. MEREC-normalized decision matrix

Alternative	C1	C2	C3	C4	C5	C6	C7	C8
A1	1.000	1.000	0.200	0.500	0.750	0.667	0.200	0.250
A2	0.600	1.000	1.000	1.000	1.000	1.000	1.000	0.250
A3	0.750	0.600	0.250	0.500	0.750	0.667	0.400	0.500
A4	0.750	0.600	0.250	0.400	0.250	0.667	0.400	0.750
A5	0.750	0.600	0.250	0.400	0.750	0.333	0.200	1.000

Note: A = Alternative; C = Criteria; MEREC = Method based on the Removal Effects of Criteria.

The largest weight was assigned to contribution per order C3, with w_3 = 0.225, indicating that this criterion produced the greatest change in aggregate alternative performance when removed from the decision structure. Logistics cost per order C7 received the second-largest weight, with w_7 = 0.195, followed by implementation complexity C8, with w_8 = 0.168, and retained order rate C4, with w_4 = 0.122. Return rate C6 and unclaimed shipments C5 received weights of w_6 = 0.087 and w_5 = 0.085, respectively, while purchase conversion rate C1 and average order value C2 received the lowest weights, at w_1 = 0.059 and w_2 = 0.058.

The overall MEREC performance measure S_i (Table 14) was subsequently calculated using Eq. (2).

The influence of removing each criterion was then determined (Table 15) using Eq. (3).

Eq. (4) was then applied to determine the total removal effect E_j (Table 16).

Table 14. Overall MEREC performance measure

Alternative	S_i
A1	0.559
A2	0.213
A3	0.499
A4	0.567
A5	0.567

Note: A = Alternative; MEREC = Method based on the Removal Effects of Criteria.

Table 15. Alternative performance after criterion removal

Alternative	C1	C2	C3	C4	C5	C6	C7	C8
A1	0.559	0.559	0.437	0.508	0.538	0.530	0.437	0.455
A2	0.160	0.213	0.213	0.213	0.213	0.215	0.213	0.062
A3	0.477	0.460	0.388	0.445	0.477	0.468	0.427	0.445
A4	0.546	0.530	0.463	0.499	0.463	0.537	0.499	0.546
A5	0.546	0.530	0.463	0.499	0.546	0.485	0.445	0.567

Note: A = Alternative; C = Criteria.

Table 16. Total criterion-removal effects

	C1	C2	C3	C4	C5	C6	C7	C8
E_j	0.116	0.113	0.440	0.239	0.167	0.171	0.383	–

Note: C = Criteria; "—" indicates that the value is not applicable to the total row.

Table 17. MEREC criterion weights

	C1	C2	C3	C4	C5	C6	C7	C8
w_j	0.059	0.058	0.225	0.122	0.085	0.087	0.195	0.168

Note: C = Criteria; MEREC = Method based on the Removal Effects of Criteria.

The final criterion weights (Table 17) were obtained using Eq. (5).

These results indicate that, within the constructed decision matrix, contribution per order, logistics cost, and implementation complexity exert the strongest influence on the evaluation of the policy alternatives. The weighting structure therefore reinforces the need to assess delivery and return policies not only in terms of customer response, but also in relation to their economic and operational consequences.

The same initial decision matrix was subsequently normalized for the MOOSRA analysis (Table 18) using Eq. (7).

Table 18. MOOSRA-normalized decision matrix

Alternative	C1	C2	C3	C4	C5	C6	C7	C8
A1	0.331	0.311	0.581	0.431	0.452	0.426	0.169	0.180
A2	0.552	0.311	0.116	0.216	0.603	0.640	0.845	0.180
A3	0.442	0.518	0.465	0.431	0.452	0.426	0.338	0.359
A4	0.442	0.518	0.465	0.539	0.151	0.426	0.338	0.539
A5	0.442	0.518	0.465	0.539	0.452	0.213	0.169	0.718

Note: A = Alternative; C = Criteria; MOOSRA = Multi-Objective Optimization on the basis of Simple Ratio Analysis.

The MEREC weights were then applied to the normalized matrix.

The values presented in Table 19 are rounded to three decimal places for reporting purposes. The final MOOSRA performance scores were calculated using the unrounded normalized values and the full-precision MEREC weights.

Because the proposed framework integrates MEREC-derived weights, the final MOOSRA values were calculated (Table 20) using the weighted formulation in Eq. (8).

Alternative A1 obtained the highest MOOSRA performance value, with Y_1 = 1.590, and was therefore ranked first. This alternative represents customer-paid delivery without a free-delivery threshold. Its leading position reflects a favorable balance between contribution per order, logistics cost, and implementation simplicity within the

constructed decision model.

Table 19. Weighted normalized MOOSRA decision matrix

Alternative	C1	C2	C3	C4	C5	C6	C7	C8
A1	0.020	0.018	0.131	0.053	0.039	0.037	0.033	0.030
A2	0.033	0.018	0.026	0.026	0.051	0.056	0.165	0.030
A3	0.026	0.030	0.104	0.053	0.039	0.037	0.066	0.060
A4	0.026	0.030	0.104	0.066	0.013	0.037	0.066	0.091
A5	0.026	0.030	0.104	0.066	0.039	0.019	0.033	0.121

Note: A = Alternative; C = Criteria; MOOSRA = Multi-Objective Optimization on the basis of Simple Ratio Analysis.

Table 20. MOOSRA performance scores and final ranking

Alternative	Y_i	Ranking
A1	1.590	1
A2	0.341	5
A3	1.055	4
A4	1.096	2
A5	1.073	3

Note: A = Alternative; MOOSRA = Multi-Objective Optimization on the basis of Simple Ratio Analysis.

Alternative A4 ranked second, with $Y_4 = 1.096$. This policy combines a free-delivery threshold with a prepayment incentive. Its relatively strong performance indicates that encouraging advance payment can partially reduce unsuccessful-delivery exposure while preserving the customer-facing advantages of threshold-based free delivery.

Alternative A5 ranked third, with $Y_5 = 1.073$. This alternative combines threshold-based free delivery with enhanced return management, including standardized packaging, transparent return procedures, and systematic recording of return reasons. Although these measures improve return-related and operational performance, the additional implementation burden reduces its overall score relative to A1 and A4.

Alternative A3 ranked fourth, with $Y_3 = 1.055$. This policy provides free delivery above RSD 3,000 but does not include either a prepayment incentive or enhanced return-management measures. Its performance therefore reflects a trade-off between customer attractiveness and the additional logistics costs associated with the delivery incentive.

Alternative A2, representing unconditional free delivery, obtained the lowest score, with $Y_2 = 0.341$, and ranked fifth. Within the modeled decision structure, the increase in customer attractiveness associated with unconditional free delivery was insufficient to compensate for its substantially higher logistics-cost exposure and weaker contribution performance.

The final ranking demonstrates that the most attractive customer-facing policy is not necessarily the most favorable overall alternative. In the present model, customer-paid delivery achieved the highest overall performance because of its strong contribution and low operational burden, while policies incorporating free-delivery incentives performed less favorably unless accompanied by measures that reduced logistics risk or improved operational control. The results therefore reinforce the need to evaluate delivery and return policies jointly across customer, financial, logistics-risk, and implementation dimensions.

5.4 Sensitivity and Robustness Analysis

Because the initial decision matrix is based on scenario-based ordinal assessments, the stability of the final ranking was examined under controlled perturbations of both the assessment scores and the MEREC criterion weights. Two complementary sensitivity tests were conducted. The first examined local uncertainty in the decision matrix, whereas the second examined the sensitivity of the MOOSRA ranking to moderate changes in individual criterion weights.

For the first test, each entry of the initial decision matrix in Table 12 was perturbed individually by one scale point, subject to the original five-point bounds. A score was therefore decreased by one where the resulting value remained at least 1 and increased by one where the resulting value remained no greater than 5. This procedure generated 65 feasible single-entry perturbation scenarios. For every scenario, the complete MEREC procedure was repeated to obtain a new set of criterion weights, after which the MOOSRA scores and rankings were recalculated (Table 21). This design evaluates whether a modest change in any single scenario-based assessment is sufficient to alter the principal decision outcome.

The local perturbation analysis confirmed the stability of the principal decision outcome. A1 remained the first-ranked alternative in all 65 perturbation scenarios, corresponding to a first-place stability rate of 100%. A2 similarly

remained fifth in every scenario. Greater variation occurred among A3, A4, and A5, whose baseline MOOSRA scores were relatively close. A4 retained second place in 39 of the 65 scenarios, whereas A5 and A3 occupied second place in 18 and 8 scenarios, respectively. The complete baseline ordering was preserved in 25 scenarios, or 38.46% of the tested perturbations.

Table 21. Ranking frequencies under 65 one-step perturbations of the initial decision matrix

Alternative	Baseline Rank	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
A1	1	65	0	0	0	0
A2	5	0	0	0	0	65
A3	4	0	8	19	38	0
A4	2	0	39	17	9	0
A5	3	0	18	29	18	0

Note: A = Alternative.

Ranking similarity was additionally evaluated using Spearman's rank correlation coefficient between each perturbed ranking and the baseline order. The mean correlation across the 65 scenarios was $\overline{p}$ = 0.892, with a median of 0.900 and a minimum of 0.600. These results indicate that the overall ranking structure remains comparatively stable, while the relative positions of the closely scored middle alternatives are more sensitive to local changes in the scenario-based assessments.

A second test examined the effect of moderate changes in the MEREC criterion weights. Each baseline weight was individually decreased by 10% and increased by 10%, while all weights were subsequently renormalized to preserve the unit-sum condition. With eight criteria and two perturbation directions, this produced 16 additional weighting scenarios. The baseline decision matrix was retained, and the MOOSRA scores and rankings were recalculated for each modified weight vector.

The ranking was even more stable under the criterion-weight perturbation test. A1 remained first, A4 remained second, and A2 remained fifth in all 16 scenarios (Table 22). The complete baseline ranking was preserved in 14 scenarios, corresponding to 87.5% of the tested weight perturbations. The only changes involved an exchange between A3 and A5 in two scenarios. The mean Spearman rank correlation with the baseline ordering was 0.988, and the minimum correlation was 0.900.

Table 22. Ranking stability under $\pm$10% perturbations of individual MEREC criterion weights

Alternative	Baseline Rank	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
A1	1	16	0	0	0	0
A2	5	0	0	0	0	16
A3	4	0	0	2	14	0
A4	2	0	16	0	0	0
A5	3	0	0	14	2	0

Note: A = Alternative; MEREC = Method based on the Removal Effects of Criteria.

Taken together, the two sensitivity tests show that the principal finding of the study is robust to moderate uncertainty in both the scenario-based assessments and the criterion weights. In particular, the first-place position of customer-paid delivery (A1) did not change under any of the 81 tested perturbation scenarios. The lower stability observed among A3, A4, and A5 reflects their relatively narrow baseline score differences and suggests that distinctions among the middle-ranked policies should be interpreted more cautiously than the separation between A1 and the remaining alternatives.

5.5 Theoretical and Managerial Implications

The study contributes to e-commerce logistics research by conceptualizing delivery and return policies as interconnected components of a broader decision problem rather than as independent operational practices. Delivery incentives can influence customer purchasing behavior while simultaneously changing fulfillment costs, unsuccessful-delivery exposure, post-delivery returns, and transaction-level contribution.

A first theoretical implication concerns the relationship between demand stimulation and economic sustainability. Conventional performance measures such as order volume and purchase conversion capture only the customer-response stage of the transaction. The present framework extends this perspective by incorporating contribution per order, retained-order performance, logistics cost, unsuccessful-delivery risk, and return behavior within the same analytical structure.

A second implication concerns the treatment of unclaimed shipments. Rather than interpreting unsuccessful deliveries solely as operational failures, the framework incorporates the unclaimed shipments as a distinct cost-oriented criterion. This reflects the fact that an unclaimed parcel may generate transportation, packaging, handling, inventory, and customer-acquisition costs without generating a completed sale.

A related contribution is the distinction between successful delivery and retained order rate. The economic outcome of an order is not necessarily finalized when the customer receives the parcel, because a subsequent return may reverse part of the revenue and generate additional reverse-logistics expenditure. The retained-order rate therefore provides a more complete representation of realized transaction performance.

From a methodological perspective, the integrated MEREC–MOOSRA framework demonstrates how objective weighting and multi-criteria ranking can be combined to support the structured evaluation of e-commerce logistics policies. MEREC determines criterion importance from the internal structure of the decision matrix, while MOOSRA provides a transparent ratio-based mechanism for comparing weighted benefits and costs. The framework therefore makes the evaluation logic explicit and facilitates interpretation of the criteria responsible for differences among policy rankings.

The robustness analysis adds an important qualification to this interpretation. Although the first-place position of A1 was unaffected by the tested perturbations, the relative ordering of A3, A4, and A5 was more sensitive because their baseline MOOSRA scores were close. The framework should therefore distinguish between a robust leading alternative and more weakly separated middle-ranked alternatives rather than treating every rank position as equally stable. From a sustainability perspective, the framework also highlights the resource implications of delivery and return policies. Unsuccessful shipments, repeated transportation, repackaging, additional handling, and reverse-logistics activities consume logistics resources without necessarily generating retained economic value. Although the present model does not quantify environmental impacts directly, reducing such avoidable flows is consistent with more resource-efficient e-commerce logistics. Sustainability in the present framework should therefore be interpreted primarily in terms of economic viability, operational efficiency, and the reduction of unnecessary logistics activity.

From a managerial perspective, the findings indicate that delivery policies should not be assessed solely through order volume or conversion. A policy can stimulate demand while simultaneously increasing delivery expenditure, unsuccessful deliveries, returns, and other variable costs. Managers should therefore evaluate policy alternatives through a combination of customer, operational, logistics-risk, and contribution-based indicators.

Within the modeled setting, customer-paid delivery achieved the strongest overall performance, reflecting its favorable contribution per order, relatively low logistics-cost exposure, and limited implementation burden. Threshold-based free-delivery policies nevertheless performed substantially better than unconditional free delivery, particularly when combined with measures such as prepayment incentives or enhanced return management. These findings indicate that delivery incentives can remain viable when they are designed to control logistics risk and cost exposure rather than being applied universally. The effectiveness of any threshold-based policy should therefore be reassessed periodically in relation to average order value, product margins, courier charges, customer purchasing behavior, and operational complexity.

The conditions under which customer-paid and threshold-based free-delivery policies may be preferable are discussed in Section 4.9. More broadly, these findings highlight the importance of aligning delivery and return policy decisions with the retailer's financial objectives, operational capabilities, and customer-related priorities. The selection of an appropriate policy should therefore reflect the specific business context rather than relying on a universally applicable solution. The threshold should be periodically reassessed against changes in average order value, product margins, courier charges, and customer purchasing behavior. For lower-value products, bundled offers or complementary product combinations can provide an alternative mechanism for increasing basket value.

The third-place performance of A5 also indicates that enhanced return management provides meaningful operational advantages, although these benefits must be weighed against additional implementation complexity. Systematic recording of return reasons, standardized packaging, and transparent return procedures can support better control of reverse-logistics risks and help identify recurring problems involving product presentation, order accuracy, supplier quality, or courier performance.

Managers should additionally distinguish among created orders, dispatched shipments, successfully delivered orders, and retained orders. Monitoring only created orders can overstate business performance because unsuccessful deliveries and subsequent returns reduce the economic value generated by the original customer acquisition.

Finally, the results reinforce the need for closer integration between marketing and logistics decisions. Promotional campaigns and delivery incentives should not be evaluated only through website traffic, conversion, or nominal order creation. Their effectiveness should also be assessed through delivery success, unclaimed shipments, returns, retained order rate, logistics expenditure, and financial contribution.

The integrated MEREC–MOOSRA framework therefore provides a structured decision-support mechanism for comparing alternative delivery and return policies before implementation. By making the underlying criteria, weights, benefit–cost structure, and ranking procedure explicit, the framework supports more transparent evaluation

of trade-offs among customer responsiveness, operational feasibility, logistics risk, and economic sustainability.

6 Conclusions

This study developed an integrated MEREC-MOOSRA decision-analytics framework for evaluating alternative delivery and return policies in sustainable e-commerce logistics. By combining customer-related, operational, logistics-risk, and financial criteria within a unified multi-criteria structure, the framework extends policy evaluation beyond individual indicators such as order volume, conversion rate, or delivery cost. The results showed that customer-paid delivery achieved the highest overall performance, followed by threshold-based free delivery combined with a prepayment incentive and threshold-based free delivery with enhanced return management, whereas unconditional free delivery obtained the lowest ranking. These findings indicate that policies designed to stimulate purchasing activity should be evaluated against the logistics costs, operational burdens, and fulfillment risks generated during and after delivery.

A central implication of the analysis is that the economic outcome of an e-commerce transaction should not be assessed at the point of order creation or even successful delivery alone. Unclaimed shipments and subsequent product returns can substantially alter the realized value of customer acquisition. Distinguishing among created orders, dispatched shipments, successfully delivered orders, and retained order rate therefore provides a more informative basis for evaluating delivery and return policies. Within the proposed framework, contribution per order received the greatest MEREC weight, followed by logistics cost per order and implementation complexity, while retained-order performance and return-related risks also contributed to the final decision structure. This weighting pattern reinforces the importance of jointly considering economic performance, logistics expenditure, operational burden, and fulfillment outcomes when evaluating alternative delivery and return policies.

From a methodological perspective, the study demonstrates how objective criterion weighting and ratio-based alternative ranking can be integrated to support transparent policy comparison. MEREC identifies the relative importance of criteria according to their removal effects, while MOOSRA aggregates weighted benefit- and cost-oriented performance into a single ranking structure. The resulting framework makes the evaluation logic explicit and provides a reproducible basis for examining trade-offs among customer responsiveness, operational feasibility, logistics risk, and financial performance. This structure is consistent with the broader objective of sustainability-oriented decision analytics, in which competing outcomes must be considered simultaneously rather than independently. In this context, the sustainability relevance of the framework lies not only in maintaining financial viability but also in supporting more resource-efficient logistics decisions by reducing avoidable fulfillment failures, repeated handling, and unnecessary reverse-logistics activity.

The sensitivity analysis further supported the stability of the principal conclusion. Customer-paid delivery (A1) remained the first-ranked alternative in all 65 one-step decision-matrix perturbations and in all 16 criterion-weight perturbation scenarios. At the same time, the relative positions of A3, A4, and A5 varied under several perturbations, indicating that the distinction among these middle-ranked alternatives is less robust than the first-place position of A1.

Several limitations should be acknowledged. First, the analysis is based on a modeled e-commerce setting rather than transaction records from a specific operating company. The numerical relationships and policy scores should therefore be interpreted as scenario-based evidence rather than as universally applicable empirical estimates. Second, the initial decision matrix was constructed from the modeled operating conditions, the practical analysis, and the relevant literature rather than from externally calibrated measurements or a large-scale expert elicitation process. Although the sensitivity analysis showed that the first-ranked alternative remained stable under the tested score and weight perturbations, this does not eliminate the uncertainty associated with the construction of the original ordinal assessments. Third, the present study addresses sustainability primarily through economic viability, operational efficiency, and the reduction of avoidable logistics activity. Environmental and broader social sustainability indicators, such as transport emissions, packaging waste, energy use, and stakeholder equity, were not included directly in the current decision structure. Fourth, the analysis represents a specific business configuration characterized by relatively low- to medium-value products, outsourced delivery, and a substantial share of cash-on-delivery transactions. The resulting rankings may therefore change under different product mixes, payment structures, logistics networks, customer markets, or cost conditions.

Future research should therefore validate the framework using real transaction-level data from e-commerce companies of different sizes, product categories, and operating models. Longitudinal applications would help determine whether policy rankings remain stable under changes in demand, courier performance, customer behavior, and market conditions. Future studies should extend the present robustness analysis by examining broader multi-parameter perturbations and by comparing the integrated MEREC-MOOSRA framework with alternative weighting and ranking methods. Extending the criterion system to incorporate customer segmentation, preference heterogeneity, transport emissions, packaging waste, energy use, and broader social indicators would allow a more comprehensive assessment of environmental, economic, and social sustainability. Finally, probabilistic, fuzzy, interval, or scenario-

based extensions could be introduced to represent uncertainty explicitly and examine how incomplete or imprecise information affects policy priorities.

Overall, the study shows that delivery and return policies are best treated as a structured multi-criteria decision problem rather than as isolated operational choices. The proposed framework provides a transparent basis for comparing alternative policies and clarifies how customer response, logistics risk, retained order rate, and economic performance interact in sustainable e-commerce decision-making.

Author Contributions

Conceptualization, A.S.B., M.A., and V.P.; methodology, A.S.B., M.A., and V.P.; software, M.A. and V.P.; writing-original draft preparation, A.S.B., M.A., and V.P.; writing-review and editing, A.S.B., M.A., and V.P. All authors have read and agreed to the published version of the manuscript.

Data Availability

The modeled data used in this study and the information required to reproduce the reported analyses are provided within the article. No confidential company-level transaction data were used.

Conflicts of Interest

The authors declare no conflicts of interest.

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